Bapatla Engineering College

Department of CSE (AIML)

Paper to Pill: AI-Driven Prescription Conversion

CM09

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PO1 PO2 PO4 PO12

Introduction & Problem Statement

Medical Document Confusion

- 65% can't understand prescriptions
- 58% confused about lab results

Language Barriers

- 81% prefer native language
- Most apps only in English

Medication Errors

- 40% errors from misunderstanding
- Risk of over/under-dosing

No AI Analysis

- Existing tools: OCR only (60-70%)
- No context understanding

Current Solutions & Issues

- Manual Consultation: Time-consuming, inconsistent
- Generic Apps: Lack specificity, often inaccurate
- Language Barriers: English-centric, excludes diverse groups

Our AI-Powered Solution

- Dual-Input Engine: Text & voice analysis (prescriptions, labs)
- Multilingual Support: 20+ languages
- Real-time Analysis: Instant interpretation, personalized guidance

Key Benefits

- 85%+ Accuracy: Reliable information
- Patient-Friendly: Easy for all demographics
- Secure & Private: HIPAA-compliant
- Accessible 24/7: Anytime, anywhere information

PO1 PO2 PO4 PO12

Objectives of the Project

1

Develop Dual-Input AI System

Achieve >85% accuracy combining Easy-OCR and Groq Vision LLM for robust analysis.

2

Provide Multilingual Assistance

Offer healthcare assistance in 5 languages: English, Telugu, Hindi, Tamil, Kannada.

3

Create Integrated Platform

A unified system for prescriptions, lab reports, and comprehensive drug guidance.

4

Implement Secure Authentication

Ensure secure user authentication and thorough medical history tracking.

5

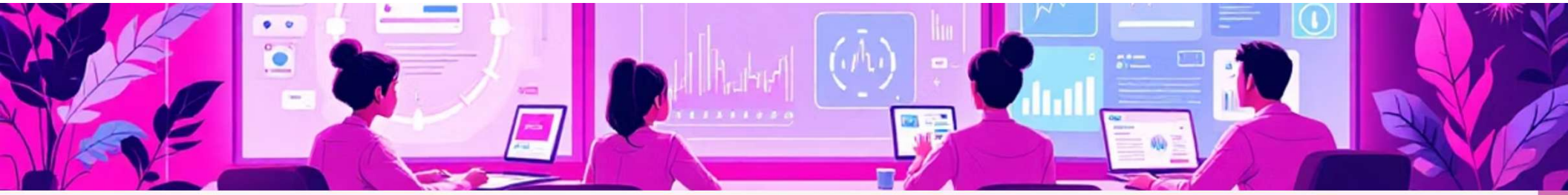
Deploy 24/7 AI Chatbot

Provide real-time medical query assistance through an accessible AI chatbot.

6

Establish Evaluation Framework

Define a rigorous framework with 4 critical metrics for continuous system assessment.



Engineering Knowledge Required

Developing an AI-powered medical analyzer demands expertise across several critical domains to ensure accuracy, security, and usability.

Machine Learning & AI

Deep learning frameworks, natural language processing (NLP), computer vision, model optimization, and large language model (LLM) fine-tuning.

Medical Domain Knowledge

Understanding medical terminology, prescription writing standards, various lab report formats, and potential drug interactions.

Backend Development

Robust API design, efficient database management (SQL/NoSQL), scalable cloud deployment strategies, and proficiency in Flask framework.

Security & Compliance

Data encryption techniques, HIPAA compliance protocols, secure password hashing, and stringent privacy protection measures.

Frontend Development

Intuitive UI/UX design principles, responsive interface development, web accessibility standards, and expertise in HTML, CSS, and JavaScript.

COMPARATIVE ANALYSIS

Analysis of Existing Methodologies

Understanding current medical document analysis methods highlights our solution's innovation:

Methodology	Strengths	Weaknesses
Traditional OCR	Basic text extraction	Low accuracy (60-70%), no context
Rule-Based Systems	Predictable, static rules	Rigid, inflexible, hard to scale
Machine Learning (NN)	Pattern recognition, higher accuracy	Extensive data needed, limited multimodal understanding
Vision Language Models (VLM)	Multimodal, context-aware analysis	Computationally intensive, complex
Our Hybrid Approach	High accuracy (85-95%), multilingual, OCR + VLM	OCR quality impacts performance

Our hybrid model fills critical gaps, offering superior accuracy, contextual understanding, and accessibility.

LITERATURE REVIEW

Limitations of Existing Systems

1

OCR-Only Tools

- Low accuracy (60-70%)
- Text-only, no context
- No semantic interpretation

2

Mobile Applications

- Require manual data entry
- Limited features
- Not patient-friendly

3

CDSS (Clinical Decision Support Systems)

- Hospital-focused
- Not patient-accessible
- Lack patient-centric explanations

These solutions fall short in multilingual support, high accuracy, and patient-friendly integration.

Literature Survey & Analysis

PO Indicators: PO2

System	Method	Accuracy	Limitation
OCR-Based Systems	Tesseract	60-70%	Can't handle handwriting
CNN Medicine Recognition	Deep Learning	78%	Needs 10K+ images
NLP Text Analysis	NER	72%	Structured text only
Mobile Health Apps	Database	N/A	Manual search only

Our Solution: Dual-Input (OCR + Vision) = 85-95% Accuracy

Implementation & Data Flow

PO Indicators: PO9 | PO10 | PO11

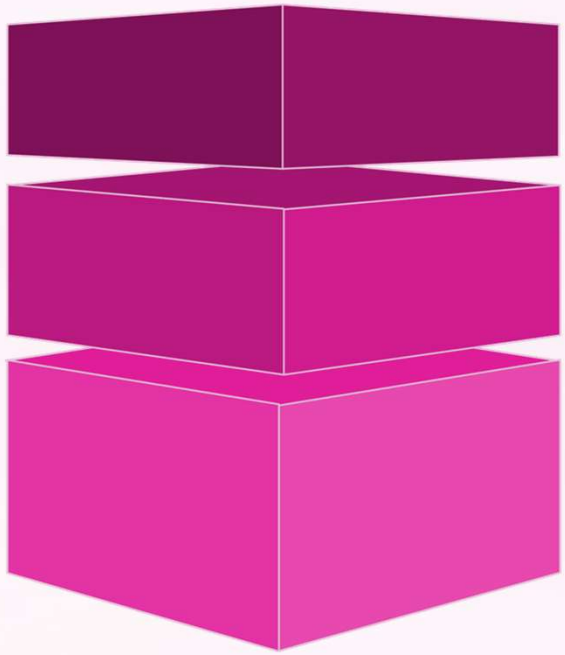
1	2
Step 1: Upload	Step 2: Validate
3	4
Step 3: OCR Extract	Step 4: Encode
5	6
Step 5: Groq API	Step 6: AI Analyze
7	8
Step 7: Save	Step 8: Display

⏱️ Processing Time: 3-5 seconds

SYSTEM ARCHITECTURE PO3

System Architecture - Three-Tier Design

Presentation LayerHTML5, CSS3, JavaScript, Responsive UI



Application LayerFlask, Python, AI Processing, Business Logic

Data LayerMongoDB, File Storage, User Data; External Groq & EasyOCR

This diagram illustrates the modular three-tier architecture, designed for scalability, maintainability, and robust data handling. Each tier has distinct responsibilities, and external services are integrated to enhance core functionalities like AI processing and text extraction.

Design Methodology & Approach

Our core design principles ensure a robust, user-friendly, and secure AI-powered prescription analyzer.

1

Requirement Analysis
Identified gaps in existing systems

2

System Design
Three-tier architecture with MVC pattern

3

Technology Selection
Groq LLM + EasyOCR + Flask + MongoDB

4

Implementation
Dual-input processing pipeline

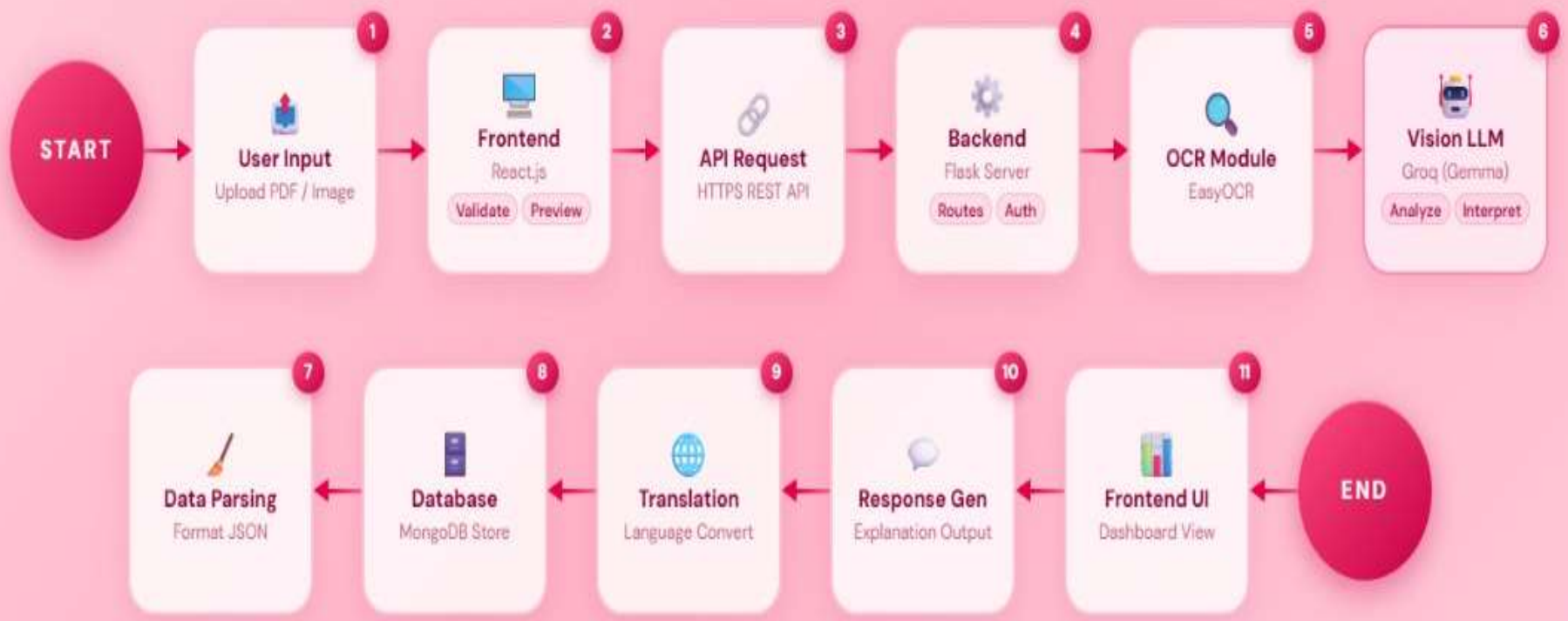
5

Testing & Evaluation
4-metric quality framework

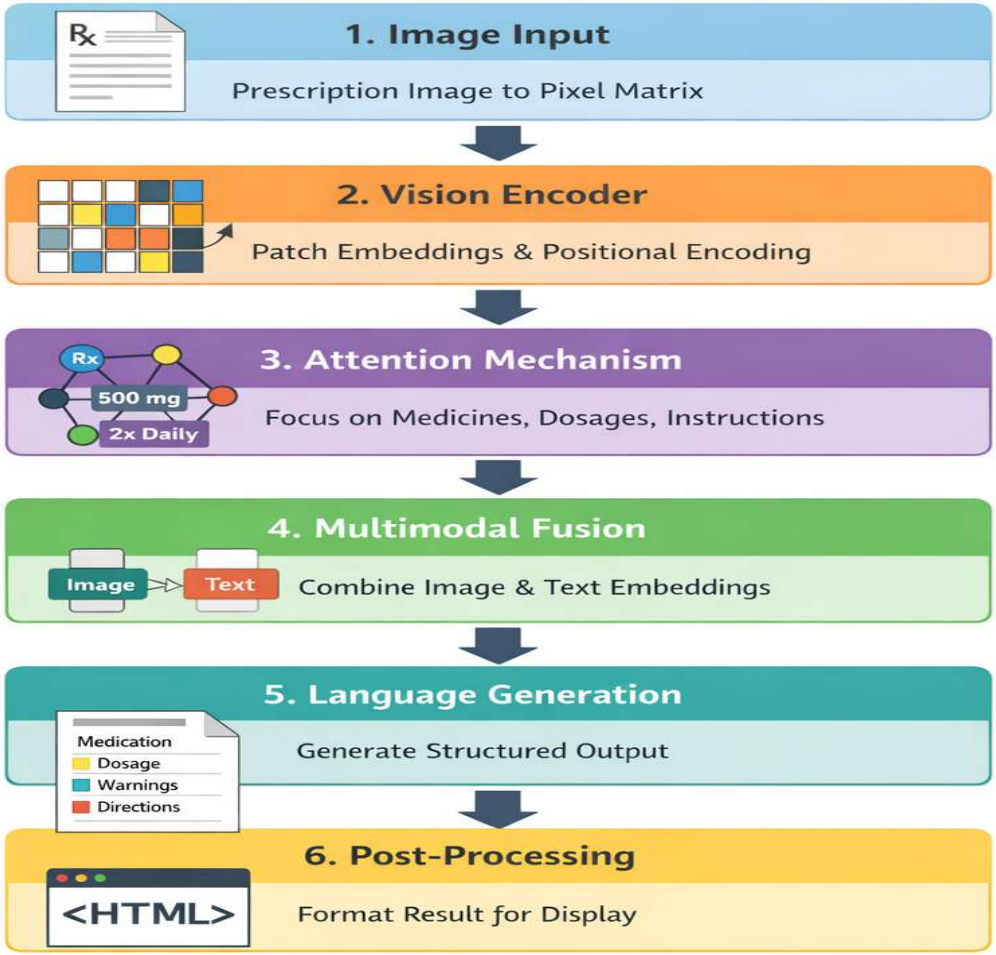
6

Deployment
Cloud-ready scalable architecture

Flow Chart


START
1
User Input
Upload PDF / Image
2
Frontend
React.js
Validate
Preview
3
API Request
HTTPS REST API
4
Backend
Flask Server
Routes
Auth
5
OCR Module
EasyOCR
6
Vision LLM
Groq (Gemma)
Analyze
Interpret
7
Data Parsing
Format JSON
8
Database
MongoDB Store
9
Translation
Language Convert
10
Response Gen
Explanation Output
11
Frontend UI
Dashboard View
END

Groq Vision LLM Processing Flow



Planning & Teamwork

01

Project Initiation
Define scope, goals, deliverables.

02

Team Formation
Assign roles (Dev, AI/ML, QA, PM).

03

Planning Phase
Create timeline, milestones, sprints.

04

Resource Allocation
Distribute tasks & responsibilities.

05

Communication Setup
Establish collaboration tools.

06

Execution
Implement features in sprints.

07

Monitoring & Feedback
Track progress, gather feedback.

08

Delivery & Handoff
Deploy system & document.

Modules in Project

Our AI medical assistant features distinct, interconnected modules, ensuring robust functionality and comprehensive patient support.

Frontend Module

User-facing interface (React) for authentication, dashboard display, and secure document uploads.

Database Module

Manages MongoDB connections for flexible data storage, history tracking, and patient data integrity.

Backend Module

Flask-powered API for route handling and processing all frontend requests efficiently.

Security Module

Implements password hashing, encryption protocols, and HIPAA compliance for sensitive patient data.

AI/ML Module

EasyOCR for text extraction and Groq Vision LLM for intelligent interpretation and model inference.

Translation Module

Provides multilingual support for English, Telugu, Hindi, Tamil, and Kannada.

PO5

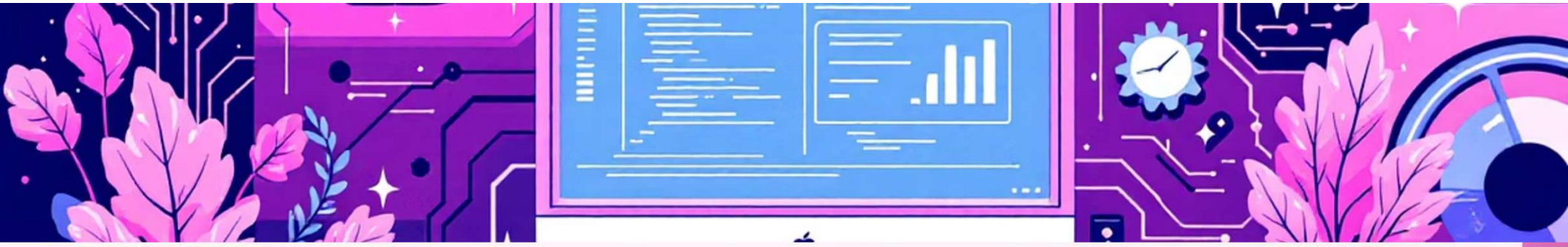
Technology Stack & Tools

Frontend
- HTML5, CSS3
- JavaScript ES6+
- Responsive Design

Backend
- Python 3.10+
- Flask 3.0.0
- Werkzeug

AI/ML
- Groq API
- EasyOCR 1.7.1
- Vision LLM

Database
- MongoDB 6.0+
- NoSQL
- Document Storage



Key Features & Innovations

1

Dual-Input Processing

OCR + Vision: 85-95% accuracy

2

Multilingual

5 Indian languages supported

3

Prescription Analyzer

Medicines, dosages, warnings

4

Lab Report Interpreter

Color-coded normal/abnormal

5

24/7 AI Chatbot

Real-time query assistance

6

Secure & Private

Password hashing, encryption

Thank you!

“Paper to Pill converts confusion into clarity and paper into safe digital healthcare.”